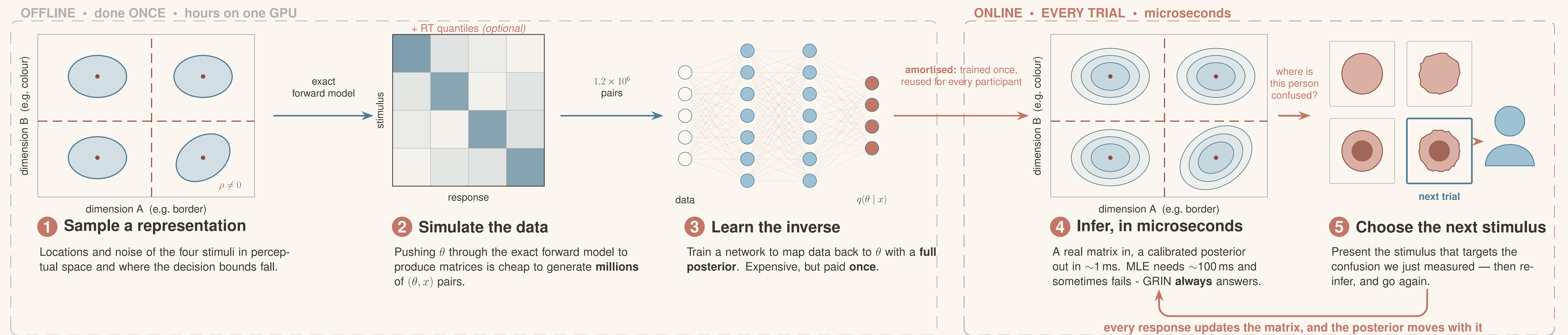


Adaptive Perceptual Training via Simulation-Based Inference of GRT Models

Fitting a perceptual model fast enough to teach with it

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What we are measuring

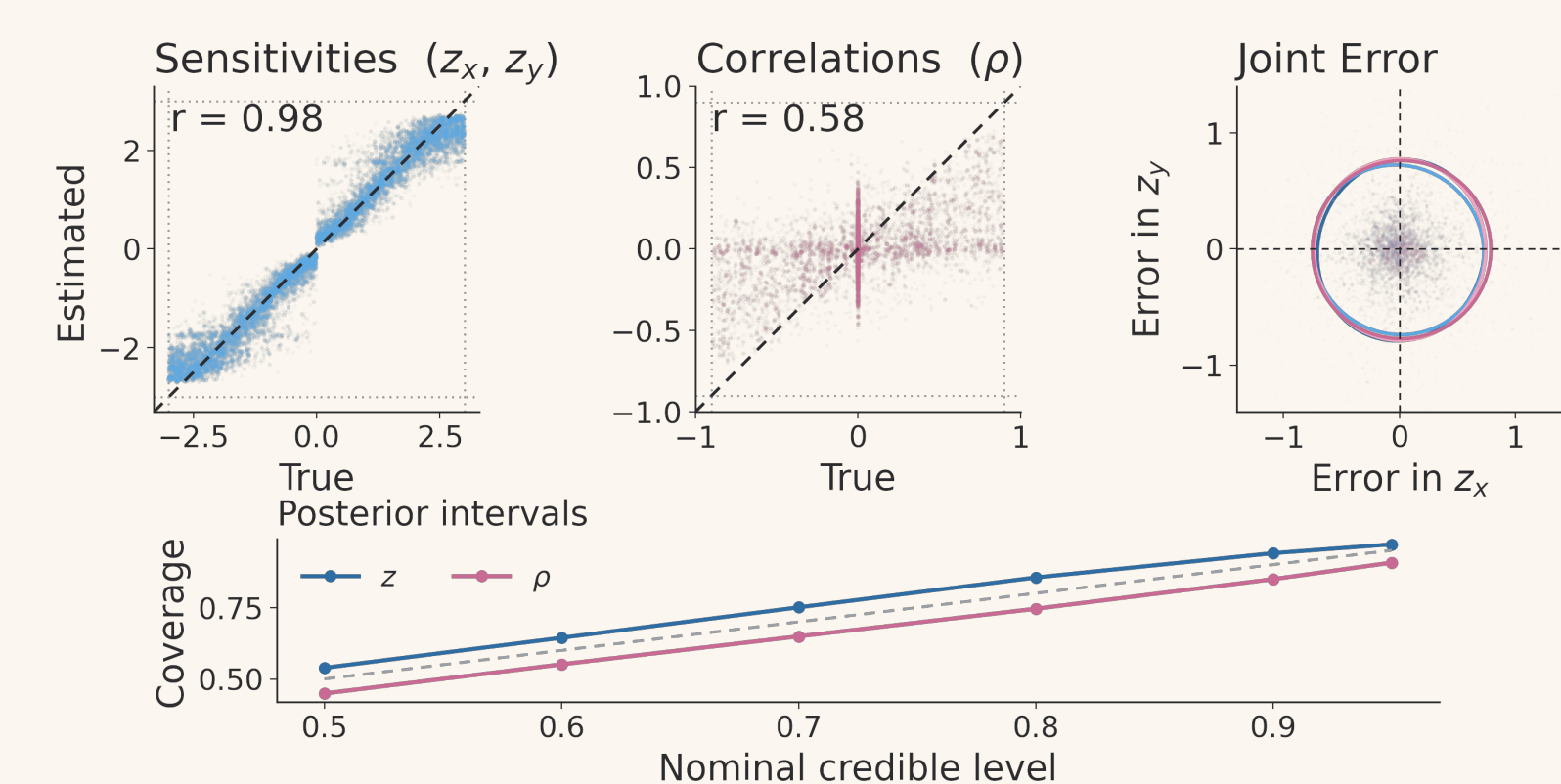
General recognition theory (GRT) describes how a person perceives and categorises a stimulus made of two features. Stimuli are placed in a perceptual space with the goal of recovering (i) their location, (ii) their noise, (iii) perceptual independence, and where the person sets their decision bounds.

In the clinic. For melanoma identification, confusing an irregular *border* with variegated *colour* is a different failure from not seeing the colour variegation at all and each needs different practice. Accuracy alone cannot tell them apart but a multidimensional representation can.

The same performance split can show up in training. Three trainees could score identically yet differ underneath: one lacks the sensitivity to see a feature, another perceives it but has drifted their criterion, and a third perceives both but cannot separate them. Critically, each requires different training.

We aim to recover the representation behind multidimensional categorisation accuracy to facilitate efficient perceptual training.

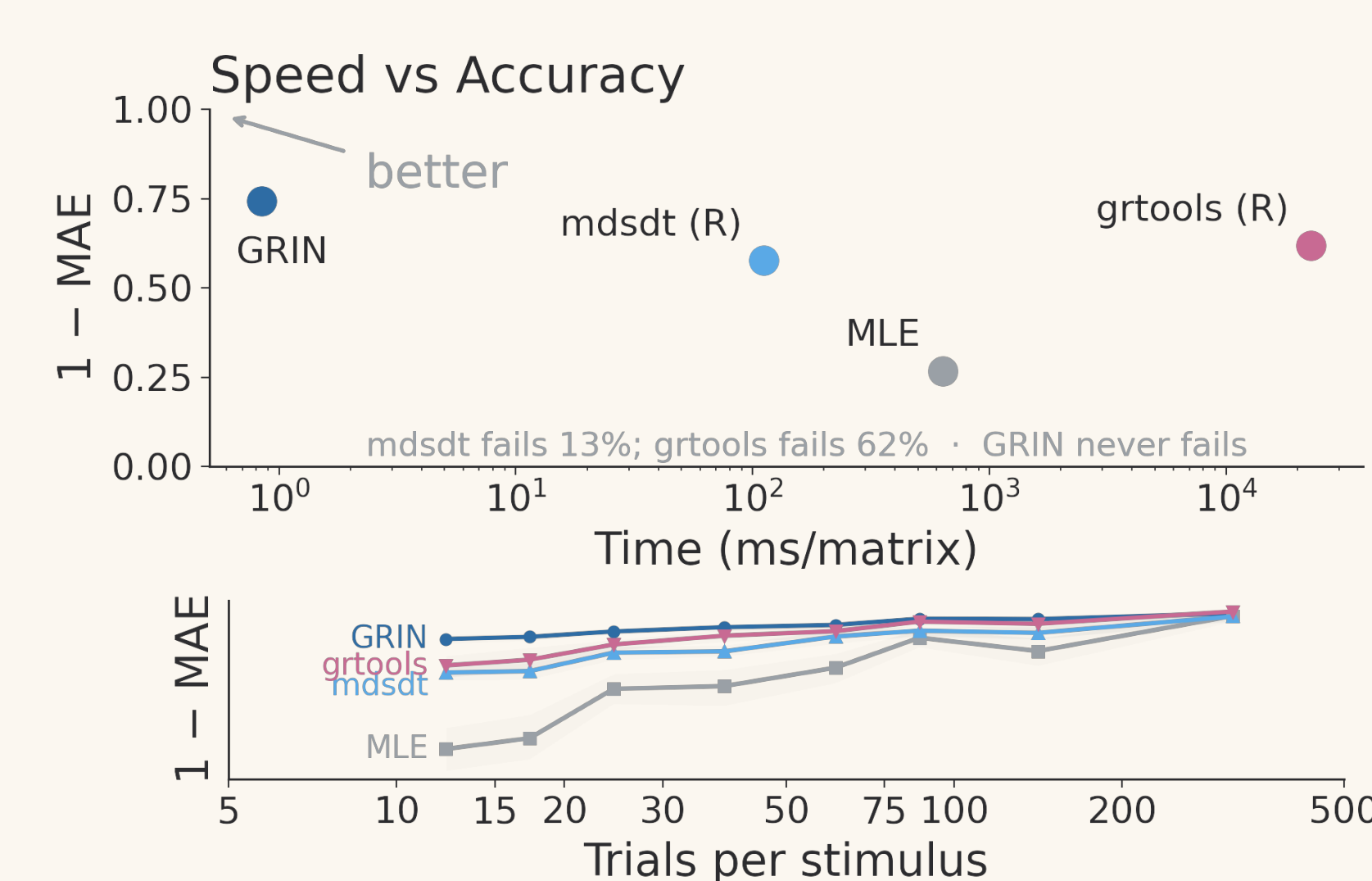
Recovery, and its limit



Recovery is uneven. Sensitivities return at $r \approx 0.975$, correlations far noisier at $r \approx 0.576$, but maximum likelihood shows the *same* split.

Weak correlations are genuinely unrecoverable from one confusion matrix, but the posterior permits **honest** credible interval coverage at their stated rate, making the method safe to act on mid-experiment.

Instant estimation



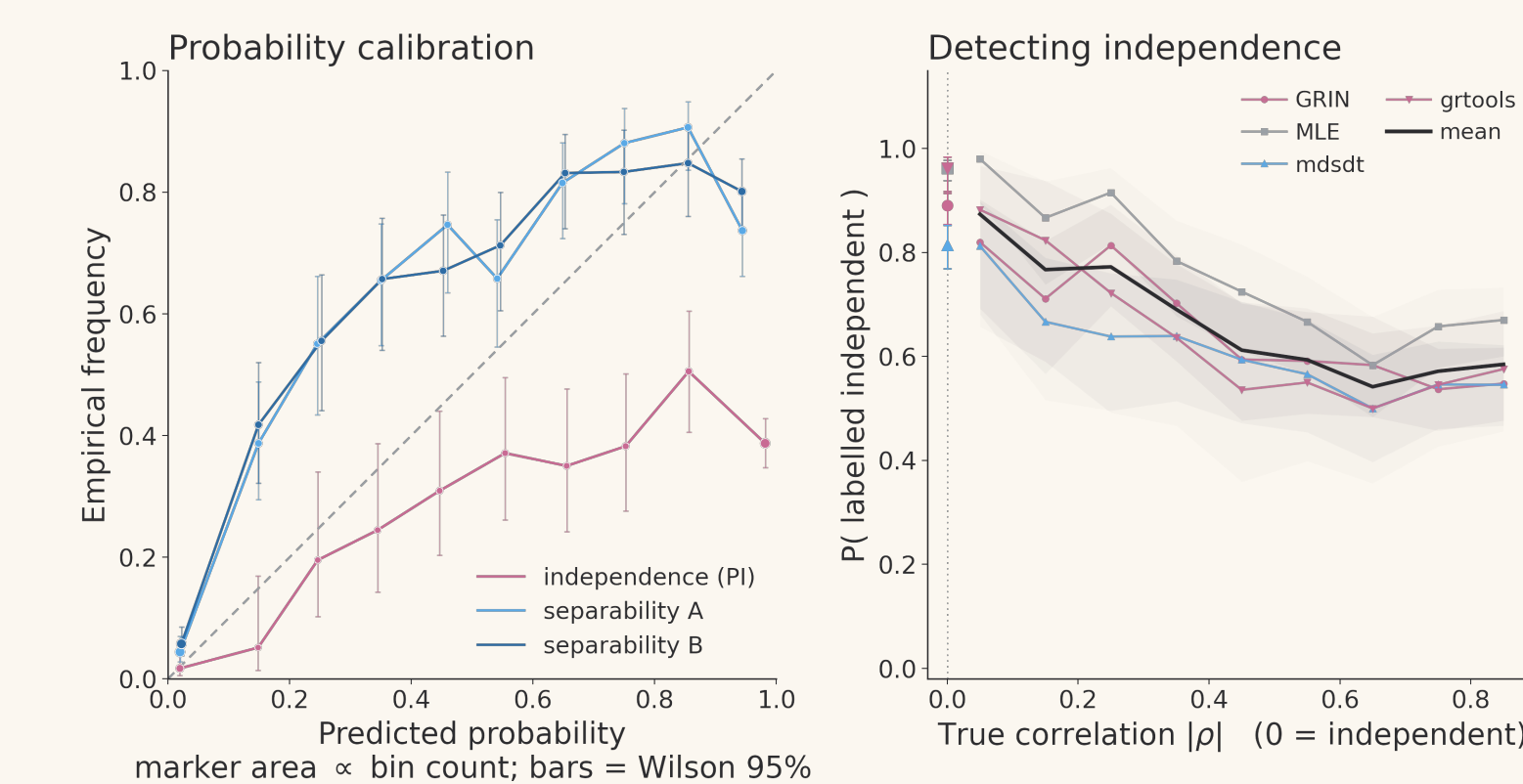
Speed x accuracy trade-off compared against our own MLE and two leading R packages (mdsdt, grtools). **Top:** fast is left, accurate is up — GRIN is both fast and accurate. Faster than all other tools and no less accurate than the R packages or Python MLE. **Bottom:** accuracy rises with sample size, and GRIN leads when data are scarce.

100x

faster than the realistic MLE workflow.

This speed wholly represents the enabling condition. At ~ 1 ms a posterior update fits inside an inter-stimulus interval, so the model can sit *inside* the trial loop and never fail to return an answer (which R and MLE fitters sometimes do).

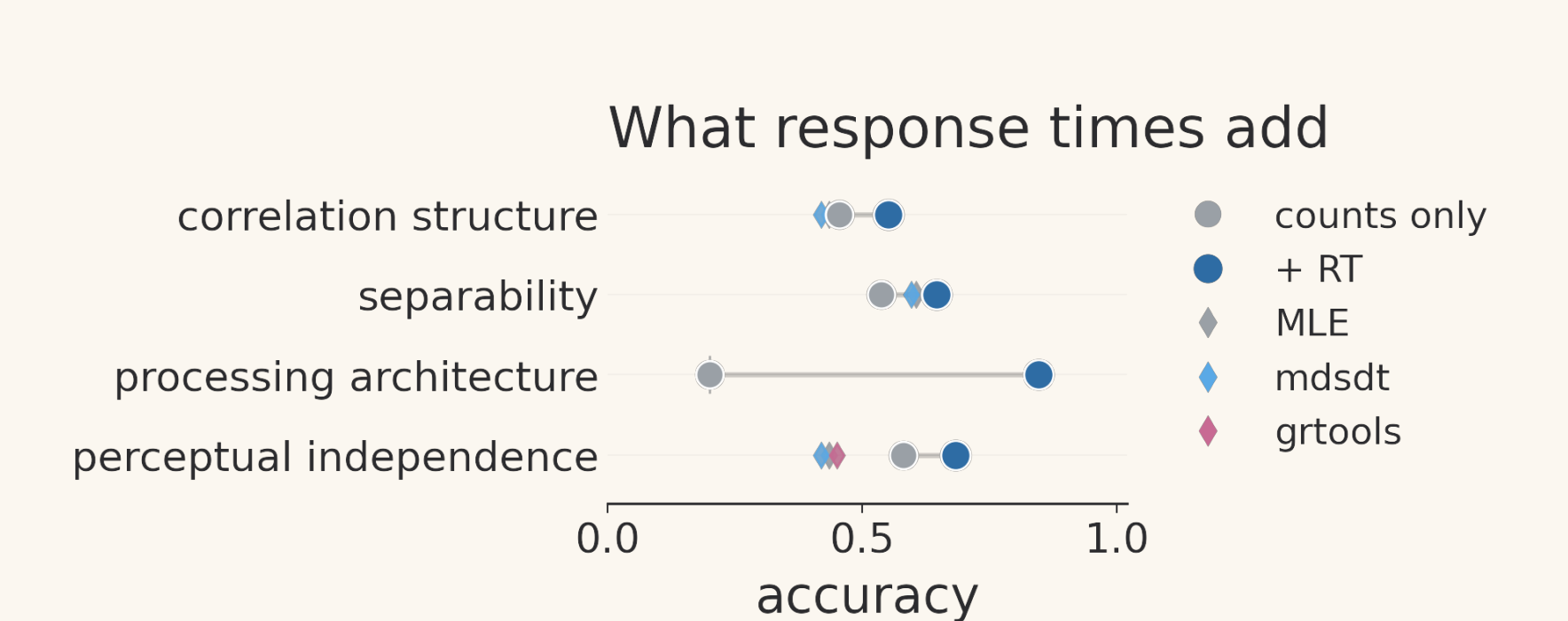
Identification



Invisible independence. Every method drifts toward signalling independence as the real correlation vanishes. Weak correlations barely affect a confusion matrix, so a challenging to identify. GRIN can help here by reporting the uncertainty where the others are committed to a point estimate.

Separability is recovered sharply but correlation structure is not. This is true for all methods, not just ours.

Implications



Timing. Response times can (mostly) recover processing architecture and dimension neglect, but is largely helpless for the correlation problem.

55%

of trials can be saved by stopping each participant at a precision threshold for model class or parameter estimate confidence instead of a fixed budget.

- Next.**
- Real-world and melanoma screening, where shape asymmetry, border irregularity, colour variance, and many more complex perceptual features genuinely overlap.
 - Develop targeted stimulus selection for trainign perceptual sensitivity and/or holistic representations.
 - One could also move from selecting stimuli in a pool to generating them.